

AUTONOMOUS CRISIS INTELLIGENCE • PRODUCT REVEAL

Rescue Drone Swarm

Mobile-deployed, AI-coordinated autonomous drone swarms engineered to find missing persons and save lives when every second counts.

The Limitations of Traditional Response



Dangerously Slow

Helicopters and ground teams face massive deployment delays.



Prohibitive Cost

Manned aircraft missions incur massive financial overhead.



High Personnel Risk

Sending human rescuers into unstable disaster zones is perilous.

Mobile Ground Command Deployment

Rapid deployment from vans and trucks directly at the disaster perimeter.

Drones are carried, charged, and launched on-site. Operators specify target search zones instantly, initiating autonomous fleet operations without waiting for central air support.



Central Messaging & Swarm Architecture



Central Board

Central messaging board informs each agent where to search.



Shared Knowledge

Live knowledge synchronization across decentralized units.

Auto-Recharge

Autonomous return-to-base and continuous rotation cycles.

Informed Starts via Satellite & DeepSearch

Intelligent Prioritisation from Earth Observation Data

By integrating satellite elevation models, population density layers, and DeepSearch coordinate analysis, the swarm starts with optimized weighting instead of blind grid searches.

Algorithm Weights

- ✓ Population density & crowd hotspots
- ✓ Hazard urgency & flood vulnerability
- ✓ Distance and battery energy cost
- ✓ Unsearched frontier maximization

Dynamic Adaptation: Flash Flood Scenario

Mission priorities shift instantly in response to unfolding environmental threats.

Example: Searching for a missing person in a forest while a flash flood approaches. The swarm dynamically reprioritizes expected flood zones to save the individual before waters rise, automatically reallocating units on the fly.



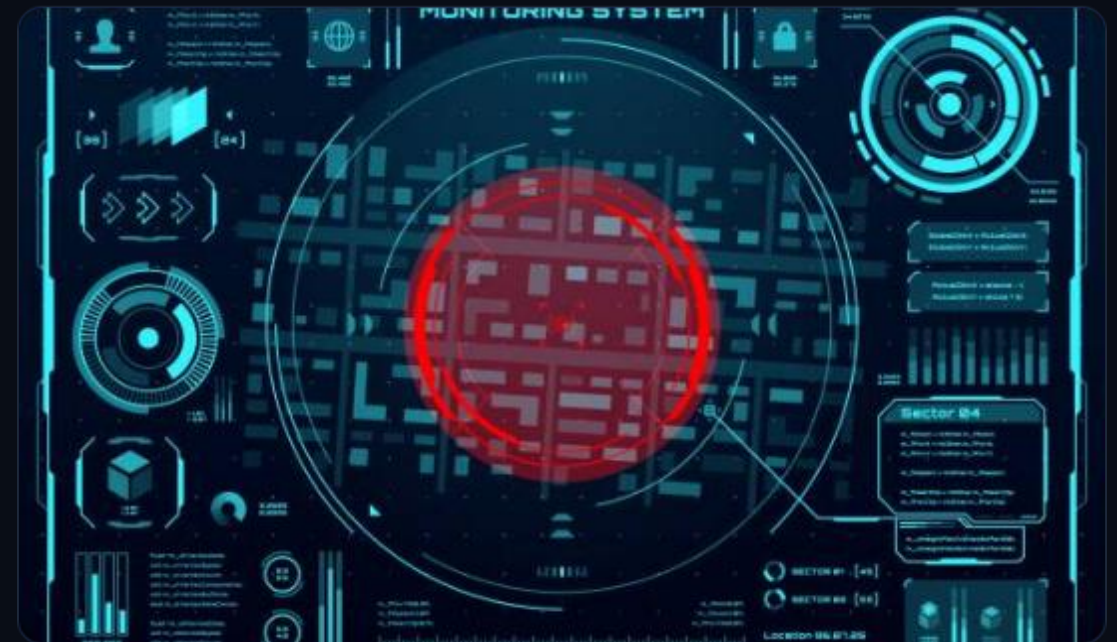
Real-Time Re-planning

Zero human intervention required to redirect drones toward high-risk temporal vectors.

Wildfire Behavior & Superior Intelligence

Drones outperform single helicopters in hazardous wildfire tracking.

When a forest fire occurs, swarms observe fire spread, smoke density, and thermal boundaries concurrently across multiple angles, yielding far richer operational data than a single manned helicopter at a fraction of the risk.



The Simulation Demo Platform



Generated Environments

Tested across diverse city and terrain area bundles.



Individual Recovery

Simulating the exact search and location of missing persons.



Live Metrics

Real-time tracking of coverage, discovery, and efficiency.

FUTURE-READY CRISIS RESPONSE

Saving Lives Through Autonomous Intelligence

Transitioning from slow, expensive manned missions to nimble, coordinated drone swarms capable of surviving and adapting to any disaster environment.
